Honors Calculus III, Homework 5

Please upload solutions to **Questions 1, 4a and 5** by **8pm on Monday, May 01**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says "optional". As usual, for full credit you need to show and explain your work.

1. (5pt) Let $X \subset \mathbf{R}^t$ be a subset. Let $f : X \to \mathbf{R}^n$ and $g : \mathbf{R}^n \to \mathbf{R}^m$ be continuous functions. Prove that $g \circ f$ is continuous.

2. Let $f : \mathbf{R}^n \to \mathbf{R}^m$ be a function. Define the graph of f by the formula
$$\Gamma_f = \{(x, y) \in \mathbf{R}^{n+m} : y = f(x)\}.$$
   (a) Assume that f is continuous. Prove that Γ_f is closed.
   (b) Prove or disprove the converse to (a).

3. (a) Let $f, g : \mathbf{R}^m \to \mathbf{R}$ be two continuous functions, and $\lambda \in \mathbf{R}$ be a number. Prove that $f + g$ is continuous, fg is continuous, and λf is continuous..
   (b) Let $P : \mathbf{R}^3 \to \mathbf{R}$ be a polynomial function of degree n: this means that there exist coefficients $\alpha_{ijk} \in \mathbf{R}$ such that
$$P(x, y, z) = \sum_{i+j+k \leq n} \alpha_{ijk} x^i y^j z^k.$$
   Deduce from part (a) that P is continuous.
   (c) Formulate and prove the analogue of part (b) for polynomial functions $P : \mathbf{R}^m \to \mathbf{R}$ for any value of m. (The only difference from (b) is that you need to come up with good notation.)

4. Assume that $X \subset \mathbf{R}^n$ is compact. This problem guides you through a proof that X is closed and bounded.
   (a) (5pt) Assume that $n = 1$. Prove that X is bounded. (Hint: if not, there is an obvious open cover of X with no finite subcover, obtained using half-infinite intervals.)
   (b) Going back to the case of general $n \geq 1$, prove that X is bounded by considering the image of X under each of the coordinate functions $c_i : \mathbf{R}^n \to \mathbf{R}$.
   (c) Let $y \in \mathbf{R}^n \setminus X$. Prove that for all $x \in X$ there exists $\epsilon_x > 0$ such that
$$B(x, \epsilon_x) \cap B(y, \epsilon_x) = \varnothing.$$
   (d) Using the compactness of X, prove that there exists $\epsilon > 0$ such that
$$B(y, \epsilon) \cap X = \varnothing.$$
   (e) Conclude that $\mathbf{R}^n \setminus X$ is open, and so X is closed.

5. (10pt) Let $X \subset \mathbf{R}^n$ be a non-compact set. Prove that there exists an unbounded continuous function

$$f : X \to \mathbf{R}.$$

(Proof sketch: argue that either X is unbounded or X is not closed (or both). In the first case, let $f(x) = ||x||$ be the norm. In the second case, let $f(x) = 1/||x - z||$ for some well-chosen point $z \in \mathbf{R}^n \setminus X$.)

6. (a) Assume that $X \subset \mathbf{R}^n$ has the property that for all $x, y \in X$ the line segment $S_{x,y}$ joining x and y is contained in X. Prove that X is connected.
   (The precise definition of $S_{x,y}$ is
   $$S_{x,y} = \{tx + (1-t)y : t \in [0,1]\}.)$$
   (b) Deduce that $\mathbf{R}^n$ is connected for all $n > 0$.
   (c) Assume that $U \subset \mathbf{R}^n$ is both open and closed. Prove that either $U = \mathbf{R}^n$ or U is empty.